Fiscal Year:	FY 2020	**Task Last Updated:**	FY 09/03/2019
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Project Title:	Microgravity Studies of Nanoparticles and Topological Defects in Liquid Crystal Thin Films		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hatch, Tyler	**Contact Phone:**	216.433.5073
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Flight Program:	ISS		
Flight Assignment:	Liquid Crystal Facility		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lacaze, Emmanuelle Ph.D. (Université Pierre et Marie Curie) Hegman, Torsten Ph.D. (Kent State University) Taylor, Philip Ph.D. (Case Western Reserve University) Lemieux, Robert Ph.D. (COLLABORATOR / University of Waterloo)		
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Task Description:	Nanoparticle / liquid crystal composites have created new avenues for scientific discovery and technological development. For example, recent studies have shown that mixtures of liquid crystals and appropriate semiconducting nanoparticles can be exploited for potential high efficiency and easily manipulatable photovoltaic devices. But the development of this and other technologies requires a fundamental understanding of how the nanoparticles incorporate and move within the liquid crystal, how they aggregate in and around topological defects, and how the nanoparticles affect the defects themselves. We have shown that fluorescent carbon dots (FCDs) may be dispersed in liquid crystals and used as a probe of defects, demonstrating that they accumulate in regions of high liquid crystal curvature so as to relieve the local strain. As a result their emitted light can serve as a quantitative indicator for the interaction and evolution of topological defects. Interpretation of these initial results, however, is compromised by the presence of substrates, which is required in an Earth-bound environment. Here we will perform experiments in the absence of a container by utilizing the Observation and Analysis of Smectic Islands In Space (OASIS) module aboard the International Space Station (ISS). By studying thin flat liquid crystal films doped with FCDs -- this will be done by dissolving the FCDs in a solvent and "shooting" slowly moving pico droplets of the mixture at the liquid crystal film -- in a microgravity environment, we will obviate the need for bounding substrates, thereby rendering a much simpler system for interpretation. In particular, we will use OASIS' cameras to study the spatial distribution and transport of the FCDs in and around topological defects in the smectic-C phase of the liquid crystal, especially close to the smectic-A phase transition temperature. A related project will be undertaken in which the film will be bombarded by droplets of methanol solvent from a pico inkjet printer involving a large mechanical impulse. The impact of the droplets will induce the formation of "islands" and "divots" of smectic liquid crystal. The goal of this part of the project will be to determine the size and thickness of these induced structures as a function of the film thickness and the droplet speed. The work will involve both experimental and theoretical components. The results will provide a clear picture of how nanoparticles distribute themselves over time, how defects in the liquid crystal interact, and how islands form on a liquid crystal film.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Liquid crystal (LC) textures and topological defects have been of interest for well over a century. Defects in two-dimensions generally are categorized in terms of the "defect strength"-- "s". Studies of defects can be used to obtain values of LC anchoring strength coefficients and elastic constants, albeit with limited accuracy. Defects are central in establishing the director profile in geometrically-confined liquid crystals, and serve as an important signature when determining the symmetry of phases, for example, distinguishing between a biaxial nematic phase and a Smectic-C (Sm-C) phase. Defect dynamics provide another important field of study, as defect motion is extremely sensitive to boundary effects and provides information about surfaces and impurities. Island and divot formation are useful in understanding the layer structure of the smectics and how layers reorganize. They provide information about the surface tension of the liquid crystal. Perhaps most important use of topological defects is that they can serve as traps for micro- and nanoparticles (NPs), as the presence of the particles reduces or eliminates the divergent energy associated with the defect core. In other words, nanoparticles (NPs) migrate within the LC in the direction of increasing stress associated with the liquid crystal's orientational order, as the presence of the particles provides relief from the large stresses that would be present at LC singularities. The incorporation of NPs into liquid crystals and their subsequent behavior in and around defects finds applications in areas such as energy harvesting, photonic structures, mirrorless lasing, micro-patterned LC devices, and chiral metamaterials for photonics and communications applications. They also may have applications in pharmaceuticals, as an alternative means of drug delivery.
Task Progress:	The third year of the project has focused on: 1. Tailoring the size and speed of the solvent droplets [methanol and isopropyl alcohol (IPA)] emanating from the ink jet printer head in order to minimize the creation of islands in the smectic film 2. Measuring the film thickness as a function of spreader speed using a prototype film holder loaned to us by Prof. Hiroshi Yokoyama 3. Determining the parameters for incident UV light on the fluorescent carbon dots 4. Examining the feasibility of adding an additional experiment in which the focus will be on the creation of islands and divots in the smectic film due to impact from a droplet emanating from the inkjet printer. Inkjet printers operate on the basis of several pulses applied to piezoelectrics, where the duration and pulse heights determine the velocity and diameter at which droplets of liquid are expelled from the aperture. We have been adjusting these parameters in order to optimize the droplet size and speed for methanol, our original choice of solvent for the carbon dots, and more recently for IPA. Using both 70 and 50 micrometers diameter aperture inkjet heads, we have been able to achieve a droplet speed of order 10 cm s-1, one to two orders of magnitude slower than the canonical speeds stated by the manufacturer. This has resulted in a decrease in the size of the islands created, an important requirement for our original project goals, that is, deposition of fluorescent carbon dots on a film and examination of the carbon dot aggregation at topological defects in the smectic-C phase and dispersal when the film is cooled into the smectic-A phase. Because the liquid crystals (9OO4 and ZA-261) chosen for the project have relatively high smectic-A phase temperatures, we have decided to do these experiments with IPA, which has a higher boiling temperature than methanol and therefore is more stable when the experiments are performed at temperatures of order 60-65oC. Recently we have added a new goal, viz., the creation of islands and divots in smectic-A films by bombardment of solvent droplets. (The solvent evaporates quickly and is chosen to have poor solubility with the liquid crystal. Thus, the island formation is due to the impulse force of the droplet.) In particular, the diameter and thickness of the islands and divots are being examined as functions of the film's original thickness and the droplet speed. We also examined a number of alternate liquid crystals that possessed the appropriate surface tensions for spreading and stability in the presence of our solvents. We eventually settled on Lemieux's group's ZA-284, which has a smectic-A temperature in the 40oC range and thus allows us to use methanol as the solvent. Lemieux has synthesized a batch of this liquid crystal for us, and it now has been incorporated into the program. Using Prof. Yokoyama's film holder, we have been drawing films at various spreader speeds in order to assess empirically how the film thickness is related to the speed. Here we have been using a spectroscopic technique with two lasers at 632 and 515 nm on reflection in order to determine the film thickness, and also measuring the reflectivity of thin films using the thickness-square technique, i.e., where the reflected intensity is approximately proportional to film

	thickness squared. We have found that there is little correlation between spreading speed and film thickness from approximately 15 nm up to 100 nm, suggesting that the actual flight experiments will need to be performed by creating a film of arbitrary thickness, measuring the thickness, performing the experiment, breaking the film, and then create another film of arbitrary thickness. To be sure, however, the thickness may be a function of the quantity of liquid crystal on the spreader, which will need to be investigated. We are in the process of examining the illumination of the quantum dots utilizing a 2W LED source at wavelength 450 nm. We have set up a lens system for collimation of the light, and currently are measuring the intensity required to detect the dots at low concentrations, i.e., the sorts of concentrations that we expect to be using in the smectic-C thin film experiments. In related projects: 1. We are continuing with our switching (“rewiring”) of in-plane line defects. Here we are attempting to use a single pair of in-plane electrodes (rather than orthogonal pairs of electrodes), coupled with a “dual frequency” liquid crystal in which the sign of the dielectric constant is a function of frequency. 2. We have demonstrated how certain types of in-plane disclination lines connecting half-integer strength surface topological defects can be switched to vertical disclinations using an electric field and a negative dielectric constant liquid crystal. On applying the field, the director is “squeezed” into the two-dimensional xy-plane, which excludes certain types of in-plane defect pairings.
Bibliography Type:	Description: (Last Updated: 12/04/2024)
Articles in Other Journals or Periodicals	Murray BS, Harkai S, Rosenblatt C, Kralj S. "Electric field driven reconfigurable multistable topological defect patterns." Nature Comm. (submitted) as of September 2019. , Sep-2019
Articles in Other Journals or Periodicals	Ferris AJ, Afghah S, Selinger RLB, Selinger JV, Rosenblatt C. "Electric field-induced crossover from 3D to 2D topological defects in a nematic liquid crystal: Experimental verification." Soft Matter (submitted) as of September 2019. , Sep-2019